

Chapter 1: Vector Analysis

Definition of a Vector

A vector is defined as a directed line segment $\overrightarrow{PQ}$ from one point (P) called the “initial point” to another point (Q) called the “terminal point.”

Vectors are denoted by bold faced letters or letters with an arrow over them.

The “magnitude” or “length” of the vector is denoted by $|\overrightarrow{PQ}|$.

Vector Algebra

1. Equal Vectors

Two vectors $\vec{A}$ and $\vec{B}$ are “equal” if they have the same magnitude and direction regardless of their initial points. Thus $\vec{A} = \vec{B}$ in the figure above.

Note: Two parallel vectors with the same magnitude and direction are equal, even if they start at different points.

2. Negative Vector

A vector having direction opposite to that of vector $\vec{A}$ but with the same magnitude is denoted by $-\vec{A}$.

Note: The negative vector has the same length as the original vector but points in the opposite direction.

Vector Addition, Subtraction, and Scalar Multiplication

3. The Sum or Resultant of Vectors

The “sum” or resultant of vectors $\vec{A}$ and $\vec{B}$ is a vector $\vec{C}$.

To find $\vec{C} = \vec{A} + \vec{B}$: - Place the initial point of vector $\vec{B}$ at the terminal point of vector $\vec{A}$ - Draw $\vec{C}$ from the initial point of $\vec{A}$ to the terminal point of $\vec{B}$

This is known as the **triangle rule** for vector addition.

4. The Difference of Vectors

The “difference” of vectors $\vec{A}$ and $\vec{B}$, denoted as $\vec{A} - \vec{B}$, is the vector $\vec{C}$ which, when added to $\vec{B}$, gives $\vec{A}$.

Alternatively, $\vec{A} - \vec{B}$ can be defined as $\vec{A} + (-\vec{B})$.

5. Multiplication of a Vector by a Scalar

Multiplication of a vector $\vec{A}$ by a scalar m produces a new vector $m\vec{A}$ whose: - **Magnitude:** $|m\vec{A}| = |m| \cdot |\vec{A}|$ (absolute value of m times the magnitude of $\vec{A}$) - **Direction:** - Same as $\vec{A}$ if $m > 0$ - Opposite to $\vec{A}$ if $m < 0$

Unit Vectors

Definition: Unit vectors are vectors having a length of one (unit length).

Finding a Unit Vector: If $\vec{A}$ is any vector with length $|\vec{A}| > 0$, then the expression $\frac{\vec{A}}{|\vec{A}|}$ represents a unit vector that has the same direction as $\vec{A}$.

Rectangular Unit Vectors and Vector Components

Rectangular Unit Vectors

The rectangular unit vectors $\vec{i}$, $\vec{j}$ and $\vec{k}$ are unit vectors having the direction of the positive x -, y - and z - axes of a rectangular coordinate system.

Note: We use right-handed rectangular coordinate systems.

Components of a Vector

Any vector can be represented with initial point at the origin (O) of a rectangular coordinate system. Let (A_1, A_2, A_3) be the coordinates of the terminal point of vector $\vec{A}$ with initial point at (O) .

The vectors $A_1\vec{i}$, $A_2\vec{j}$ and $A_3\vec{k}$ are called the “component vectors” of $\vec{A}$ in the x -, y - and z - directions respectively.

We write:

$$\vec{A} = A_1\vec{i} + A_2\vec{j} + A_3\vec{k}$$

The magnitude of $\vec{A}$ is:

$$|\vec{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$$

Position Vector, Dot Product, and Cross Product

Position Vector

The “position vector” $\vec{r}$ from (O) to the point (x, y, z) is written:

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

and has magnitude:

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

Dot or Scalar Product

Definition:

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta, \quad 0 \leq \theta \leq \pi$$

where θ is the angle between vectors $\vec{A}$ and $\vec{B}$.

Property: $\vec{A} \cdot \vec{B}$ is a scalar and not a vector.

Component Form: If $\vec{A} = A_1\vec{i} + A_2\vec{j} + A_3\vec{k}$ and $\vec{B} = B_1\vec{i} + B_2\vec{j} + B_3\vec{k}$, then:

$$\vec{A} \cdot \vec{B} = A_1B_1 + A_2B_2 + A_3B_3$$

Dot Product with Itself:

$$\vec{A} \cdot \vec{A} = A_1^2 + A_2^2 + A_3^2$$

$$\vec{B} \cdot \vec{B} = B_1^2 + B_2^2 + B_3^2$$

Cross or Vector Product

Definition:

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta \vec{u}, \quad 0 \leq \theta \leq \pi$$

where $\vec{u}$ is a unit vector.

Property: $\vec{A} \times \vec{B}$ is a vector whose direction is indicated by the unit vector $\vec{u}$ which is perpendicular to the plane of $\vec{A}$ and $\vec{B}$, and $\vec{A}$, $\vec{B}$ and $\vec{u}$ form a right-handed system.

Cross Product as a Determinant

Cross Product Formula

If:

$$\vec{A} = A_1\vec{i} + A_2\vec{j} + A_3\vec{k}$$

and:

$$\vec{B} = B_1\vec{i} + B_2\vec{j} + B_3\vec{k}$$

Then:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

This determinant can be expanded as:

$$\vec{A} \times \vec{B} = \vec{i} \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \vec{j} \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \vec{k} \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix}$$

$$= (A_2B_3 - A_3B_2)\vec{i} - (A_1B_3 - A_3B_1)\vec{j} + (A_1B_2 - A_2B_1)\vec{k}$$

$$= (A_2B_3 - A_3B_2)\vec{i} + (A_3B_1 - A_1B_3)\vec{j} + (A_1B_2 - A_2B_1)\vec{k}$$

Vector Functions

Definition

If corresponding to each value of a scalar (u) we associate a vector $(\vec{A})$, then $(\vec{A})$ is called a “function of (u) ” denoted by $(\vec{A}(u))$.

We can write:

$$\vec{A}(u) = A_1(u)\vec{i} + A_2(u)\vec{j} + A_3(u)\vec{k}$$

Extension of Function Concept

The function concept is easily extended. Thus, if to each point $((x, y, z))$ there corresponds a vector $(\vec{A})$, then $(\vec{A})$ is a function of $((x, y, z))$ indicated by:

$$\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$$

Vector and Scalar Fields

A vector function $\vec{A}(x, y, z)$ defines a **vector field** since it associates a vector with each point of a region.

Similarly, $\phi(x, y, z)$ defines a **scalar field** since it associates a scalar with each point of a region.

Limits, Continuity and Derivatives of Vector Functions

1. Continuity

The vector function $\vec{A}(u)$ is said to be **continuous** at u_0 if given any positive number ϵ , we can find some positive number δ such that:

$$|\vec{A}(u) - \vec{A}(u_0)| < \epsilon \quad \text{whenever} \quad |u - u_0| < \delta$$

This is equivalent to the statement:

$$\lim_{u \rightarrow u_0} \vec{A}(u) = \vec{A}(u_0)$$

Derivatives and Differentials of Vector Functions

2. The Derivative of a Vector Function

The derivative of $\vec{A}(u)$ is defined as:

$$\frac{d\vec{A}}{du} = \lim_{\Delta u \rightarrow 0} \frac{\vec{A}(u + \Delta u) - \vec{A}(u)}{\Delta u}$$

provided this limit exists.

Component Form: In case $\vec{A}(u) = A_1(u)\vec{i} + A_2(u)\vec{j} + A_3(u)\vec{k}$, then:

$$\frac{d\vec{A}}{du} = \frac{dA_1}{du}\vec{i} + \frac{dA_2}{du}\vec{j} + \frac{dA_3}{du}\vec{k}$$

3. The Differential of a Vector Function

If $\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$, then:

$$d\vec{A} = \frac{\partial \vec{A}}{\partial x} dx + \frac{\partial \vec{A}}{\partial y} dy + \frac{\partial \vec{A}}{\partial z} dz$$

is the **differential** of $\vec{A}$.

4. Derivatives of Products

Derivatives of products obey rules similar to those for scalar functions. However, when cross products are involved the order may be important.

Product Rules:

(a) Scalar times vector: $\frac{d}{du}(\phi \vec{A}) = \phi \frac{d\vec{A}}{du} + \frac{d\phi}{du} \vec{A}$

(b) Dot product: $\frac{\partial}{\partial y}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{\partial \vec{B}}{\partial y} + \frac{\partial \vec{A}}{\partial y} \cdot \vec{B}$

(c) Cross product: $\frac{\partial}{\partial z}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{\partial \vec{B}}{\partial z} + \frac{\partial \vec{A}}{\partial z} \times \vec{B}$

Example: Product Rule for Dot Product

Problem

If $\vec{A}$ and $\vec{B}$ are differentiable functions of a scalar u , prove:

$$\frac{d}{du}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{du} + \frac{d\vec{A}}{du} \cdot \vec{B}$$

Method (1): Using the Definition of the Derivative

$$\frac{d}{du}(\vec{A} \cdot \vec{B}) = \lim_{\Delta u \rightarrow 0} \frac{(\vec{A} + \Delta \vec{A}) \cdot (\vec{B} + \Delta \vec{B}) - \vec{A} \cdot \vec{B}}{\Delta u}$$

Expanding the numerator:

$$= \lim_{\Delta u \rightarrow 0} \frac{\vec{A} \cdot \Delta \vec{B} + \Delta \vec{A} \cdot \vec{B} + \Delta \vec{A} \cdot \Delta \vec{B}}{\Delta u}$$

Separating terms and applying the limit:

$$= \lim_{\Delta u \rightarrow 0} \frac{\vec{A} \cdot \Delta \vec{B}}{\Delta u} + \lim_{\Delta u \rightarrow 0} \frac{\Delta \vec{A} \cdot \vec{B}}{\Delta u} + \lim_{\Delta u \rightarrow 0} \frac{\Delta \vec{A} \cdot \Delta \vec{B}}{\Delta u}$$

Evaluating the limits:

$$= \vec{A} \cdot \frac{d\vec{B}}{du} + \frac{d\vec{A}}{du} \cdot \vec{B}$$

Method (2): Using Component Form

Let $\vec{A} = A_1\vec{i} + A_2\vec{j} + A_3\vec{k}$ and $\vec{B} = B_1\vec{i} + B_2\vec{j} + B_3\vec{k}$.

Then:

$$\frac{d}{du}(\vec{A} \cdot \vec{B}) = \frac{d}{du}(A_1B_1 + A_2B_2 + A_3B_3)$$

Applying the product rule for scalar functions:

$$= \left(A_1 \frac{dB_1}{du} + A_2 \frac{dB_2}{du} + A_3 \frac{dB_3}{du} \right) + \left(\frac{dA_1}{du} B_1 + \frac{dA_2}{du} B_2 + \frac{dA_3}{du} B_3 \right)$$

Regrouping into vector dot products:

$$= \vec{A} \cdot \frac{d\vec{B}}{du} + \frac{d\vec{A}}{du} \cdot \vec{B}$$

Geometric Interpretation of a Vector Derivative

Position Vector and Space Curves

If $\vec{r}$ is the vector joining the origin O of a coordinate system and the point (x, y, z) , then specification of the vector function $\vec{r}(u)$ defines x , y and z as functions of u .

As u changes, the terminal point of $\vec{r}$ describes a **space curve** having parametric equations:

$$[x = x(u), \quad y = y(u), \quad z = z(u)]$$

Unit Tangent Vector

If the parameter u is the arc length s measured from some fixed point on the curve, then:

$$\left[\frac{d\vec{r}}{ds} = \frac{dx}{ds}\vec{i} + \frac{dy}{ds}\vec{j} + \frac{dz}{ds}\vec{k} = \vec{T} \right]$$

$\vec{T}$ is a unit vector in the direction of the tangent to the curve and is called the **unit tangent vector**.

Velocity and Acceleration

If u is the time t , then:

- $\left(\frac{d\vec{r}}{dt} \right)$ represents the **velocity** $\vec{v}$ with which the terminal point of $\vec{r}$ describes the curve.
- $\left(\frac{d^2\vec{r}}{dt^2} \right)$ represents its **acceleration** $\vec{a}$ along the curve.

Displacement Vector

For a small change Δu in the parameter:

$$[\Delta\vec{r} = \vec{r}(u + \Delta u) - \vec{r}(u)]$$

This represents the displacement vector between two points on the curve, where Δs is the corresponding arc length along the curve.

Gradient, Divergence and Curl

The Del Operator (∇)

The vector differential operator ∇ (del), is defined by:

$$\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$$

This vector operator possesses properties analogous to those of ordinary vectors. It is useful in defining three quantities which arise in practical applications and are known as the **gradient**, the **divergence** and the **curl**.

1. Gradient

Let $\phi(x, y, z)$ be defined and differentiable at each point (x, y, z) in a certain region of space (i.e., ϕ defines a differentiable scalar field). Then the **gradient** of ϕ is defined by:

$$\text{grad } \phi = \nabla \phi = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \phi$$

Expanding:

$$\nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

Interpretation of Gradient

An interesting interpretation is that if $\phi(x, y, z) = c$ is the equation of a surface, then $\nabla \phi$ is a normal to this surface.

Divergence and Curl

2. Divergence

If $\vec{A}(x, y, z) = A_1\vec{i} + A_2\vec{j} + A_3\vec{k}$ is a vector field defined and differentiable at each point $((x, y, z))$ in a certain region of space, then the **divergence** of $(\vec{A})$ is defined by:

$$\text{div } \vec{A} = \nabla \cdot \vec{A}$$

Expanding the dot product:

$$\nabla \cdot \vec{A} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot (A_1\vec{i} + A_2\vec{j} + A_3\vec{k})$$

$$= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z}$$

3. Curl

If $\vec{A}(x, y, z)$ is a differentiable vector field, then the **curl** of $(\vec{A})$ is defined by:

$$\text{curl } \vec{A} = \nabla \times \vec{A}$$

Expanding the cross product:

$$\nabla \times \vec{A} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \times (A_1\vec{i} + A_2\vec{j} + A_3\vec{k})$$

This can be written as a determinant:

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

Expanding the determinant:

$$\nabla \times \vec{A} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \vec{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \vec{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \vec{k}$$

$$\frac{\partial A_3}{\partial x} \vec{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \vec{k}]$$

Important Note: In the expansion of the determinant, the operators $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$, $\frac{\partial}{\partial z}$ must precede (A_1) , (A_2) , (A_3) .

Formulas Involving ∇

If the partial derivatives of $(\vec{A})$, $(\vec{B})$, (u) and (V) are assumed to exist, then:

1. Gradient of Sum

$$\nabla(u + V) = \nabla u + \nabla V$$

or

$$\text{grad}(u + V) = \text{grad } u + \text{grad } V$$

2. Divergence of Sum

$$\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}]$$

or

$$\text{div}(\vec{A} + \vec{B}) = \text{div } \vec{A} + \text{div } \vec{B}]$$

3. Curl of Sum

$$\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}]$$

or

$$\text{curl}(\vec{A} + \vec{B}) = \text{curl } \vec{A} + \text{curl } \vec{B}]$$

4. Divergence of Scalar Times Vector

$$\nabla \cdot (u\vec{A}) = (\nabla u) \cdot \vec{A} + u(\nabla \cdot \vec{A})]$$

5. Curl of Scalar Times Vector

$$\nabla \times (u \vec{A}) = (\nabla u) \times \vec{A} + u(\nabla \times \vec{A})$$

6. Divergence of Cross Product

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

7. Laplacian

$$\nabla \cdot (\nabla u) = \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

is called the **Laplacian** of (u) , and $(\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$ is called the **Laplacian operator**.

8. Curl of Gradient

$$\nabla \times (\nabla u) = 0$$

The curl of the gradient of (u) is zero.

9. Divergence of Curl

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

The divergence of the curl of $(\vec{A})$ is zero.

10. Curl of Curl

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

Example 1: Gradient, Divergence, and Curl

Problem

If $\phi = x^2yz^3$ and $\vec{A} = xz\vec{i} - y^2\vec{j} + 2x^2y\vec{k}$, find:

1. $\nabla\phi$
2. $\nabla \cdot \vec{A}$
3. $\nabla \times \vec{A}$
4. $\text{div}(\phi\vec{A})$
5. $\text{curl}(\phi\vec{A})$

Solution

Part (a): Gradient of ϕ

$$\nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\phi$$

$$= \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}$$

Substituting $\phi = x^2yz^3$:

$$= \frac{\partial}{\partial x}(x^2yz^3)\vec{i} + \frac{\partial}{\partial y}(x^2yz^3)\vec{j} + \frac{\partial}{\partial z}(x^2yz^3)\vec{k}$$

$$= 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$$

Part (b): Divergence of $\vec{A}$

$$\nabla \cdot \vec{A} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \cdot (xz\vec{i} - y^2\vec{j} + 2x^2y\vec{k})$$

$$= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(2x^2y)$$

$$= z - 2y + 0 = z - 2y$$

Part (c): Curl of A

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & -y^2 & 2x^2y \end{vmatrix}$$

Expanding the determinant:

$$= \left(\frac{\partial}{\partial y}(2x^2y) - \frac{\partial}{\partial z}(-y^2) \right) \vec{i} - \left(\frac{\partial}{\partial x}(2x^2y) - \frac{\partial}{\partial z}(xz) \right) \vec{j} + \left(\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial y}(xz) \right) \vec{k}$$

$$= (2x^2 - 0)\vec{i} - (4xy - x)\vec{j} + (0 - 0)\vec{k}$$

$$= 2x^2\vec{i} + (x - 4xy)\vec{j}$$

Example 1 (continued): Divergence and Curl of ϕA

Part (d): Divergence of ϕA

First, compute $(\phi \vec{A})$:

$$(\phi \vec{A}) = (x^2yz^3)(xz\vec{i} - y^2\vec{j} + 2x^2y\vec{k})$$

$$= x^3y^4\vec{i} - x^2y^3z^3\vec{j} + 2x^4y^2z^3\vec{k}$$

Now compute the divergence:

$$\text{div}(\phi \vec{A}) = \nabla \cdot (\phi \vec{A})$$

$$= \nabla \cdot (x^3y^4\vec{i} - x^2y^3z^3\vec{j} + 2x^4y^2z^3\vec{k})$$

$$= \frac{\partial}{\partial x}(x^3y^4) + \frac{\partial}{\partial y}(-x^2y^3z^3) + \frac{\partial}{\partial z}(2x^4y^2z^3)$$

$$= 3x^2y^4 - 3x^2y^2z^3 + 6x^4y^2z^2$$

Part (e): Curl of $\phi \mathbf{A}$

$$\nabla \times (\phi \mathbf{A}) = \nabla \phi \times \mathbf{A} + \phi \nabla \times \mathbf{A}$$

$$= \nabla \phi \times (x^3y^4\mathbf{i} - x^2y^3z^3\mathbf{j} + 2x^4y^2z^3\mathbf{k})$$

Using the determinant form:

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3y^4 & -x^2y^3z^3 & 2x^4y^2z^3 \end{vmatrix}$$

Expanding:

$$= \left(\frac{\partial}{\partial y}(2x^4y^2z^3) - \frac{\partial}{\partial z}(-x^2y^3z^3) \right) \mathbf{i}$$

$$+ \left(\frac{\partial}{\partial z}(x^3y^4) - \frac{\partial}{\partial x}(2x^4y^2z^3) \right) \mathbf{j}$$

$$+ \left(\frac{\partial}{\partial x}(-x^2y^3z^3) - \frac{\partial}{\partial y}(x^3y^4) \right) \mathbf{k}$$

Computing the partial derivatives:

$$= (4x^4yz^3 + 3x^2y^3z^2)\mathbf{i} + (4x^3y^2z^3 - 8x^3y^2z^3)\mathbf{j} - (2xy^3z^3 + x^3z^4)\mathbf{k}$$

Simplifying:

$$= (4x^4yz^3 + 3x^2y^3z^2)\mathbf{i} - 4x^3y^2z^3\mathbf{j} - (2xy^3z^3 + x^3z^4)\mathbf{k}$$

Example 2: Gradient is Perpendicular to Level Surfaces

Problem

Prove that $\nabla \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$, where c is a constant.

Solution

Let $(\vec{r} = x\vec{i} + y\vec{j} + z\vec{k})$ be the position vector to any point $(P(x, y, z))$ on the surface.

Then $(d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k})$ lies in the plane tangent to the surface at (P) .

Since $(\phi(x, y, z) = c)$ is constant on the surface, the total differential of (ϕ) is zero:

$$[d\phi = \frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy + \frac{\partial \phi}{\partial z}dz = 0]$$

This can be written as a dot product:

$$[(\frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k}) = 0]$$

Recognizing that the first term is $(\nabla \phi)$ and the second is $(d\vec{r})$:

$$[\nabla \phi \cdot d\vec{r} = 0]$$

Since $(d\vec{r})$ lies in the tangent plane to the surface, and $(\nabla \phi \cdot d\vec{r} = 0)$, it follows that $(\nabla \phi)$ is perpendicular to $(d\vec{r})$ and therefore perpendicular to the surface.

Conclusion

The gradient $(\nabla \phi)$ is a vector perpendicular (normal) to the level surface $(\phi(x, y, z) = c)$.

Laplacian Operator

Definition

The Laplacian of a scalar field (ϕ) is defined as:

$$[\nabla \cdot \nabla \phi = (\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}) \cdot$$

$$\left(\frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}\right)$$

Expanding the dot product:

$$= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} \right)$$

Simplifying:

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Factoring out ϕ :

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi = \nabla^2 \phi$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called the **Laplacian operator**.

Physical Interpretation of Divergence

Introduction

To give a physical interpretation for the divergence:

Consider a typical volume element in a region filled with a moving fluid.

Let $\vec{v} = v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}$ be a vector function representing at each point the instantaneous velocity of the particle of fluid.

The Loss of Fluid Through Each Face

For a rectangular volume element with dimensions (Δx) , (Δy) , (Δz) centered at point $((x, y, z))$:

Right face (positive y-direction): $[Pv_2 + \frac{\partial(Pv_2)}{\partial y} \Delta y] \Delta x \Delta z \Delta t$

Left face (negative y-direction): $[-P_{v_2} \Delta x \Delta z \Delta t]$

(Note: $\text{Density} \times \text{Volume} = \text{Mass}$)

Front face (positive x-direction): $[[P_{v_1} + \frac{\partial(P_{v_1})}{\partial x} \Delta x] \Delta y \Delta z \Delta t]$

Rear face (negative x-direction): $[-P_{v_1} \Delta y \Delta z \Delta t]$

Top face (positive z-direction): $[[P_{v_3} + \frac{\partial(P_{v_3})}{\partial z} \Delta z] \Delta x \Delta y \Delta t]$

Bottom face (negative z-direction): $[-P_{v_3} \Delta x \Delta y \Delta t]$

where (P) is the density and (Δt) is a time interval.

Equation of Continuity

Rate of Loss Per Unit Volume

Adding the contributions from all faces and dividing by $(\Delta V \Delta t = \Delta x \Delta y \Delta z \Delta t)$, we obtain in the limit:

Rate of loss per unit volume:

$$\left[-\left(\frac{\partial(P_{v_1})}{\partial x} + \frac{\partial(P_{v_2})}{\partial y} + \frac{\partial(P_{v_3})}{\partial z} \right) \right]$$

which is the divergence of the vector $(P\vec{v})$:

$$[-\nabla \cdot (P\vec{v})]$$

Physical Interpretation

Therefore, one possible interpretation of the divergence is **the rate of loss of fluid per unit volume**.

Equation of Continuity for Incompressible Fluids

For an incompressible fluid, the density (P) is constant, and there is neither gain nor loss in a general element. Hence:

$$\nabla \cdot (P \vec{v}) = P \nabla \cdot \vec{v} = 0$$

which simplifies to:

$$\nabla \cdot \vec{v} = 0$$

This is known as the **equation of continuity** for incompressible fluids.

Physical Interpretation of Curl

Problem

To interpret the curl physically:

If $\vec{v} = \vec{\omega} \times \vec{r}$, where: - $\vec{v}$ is the linear velocity vector - $\vec{\omega}$ is the angular velocity vector

Derivation

Then:

$$\text{curl } \vec{v} = \nabla \times \vec{v} = \nabla \times (\vec{\omega} \times \vec{r})$$

First, compute $\vec{\omega} \times \vec{r}$:

$$\vec{\omega} \times \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \omega_1 & \omega_2 & \omega_3 \\ x & y & z \end{vmatrix}$$

$$= (\omega_2 z - \omega_3 y)\vec{i} + (\omega_3 x - \omega_1 z)\vec{j} + (\omega_1 y - \omega_2 x)\vec{k}$$

Now compute the curl:

$$\nabla \times (\vec{\omega} \times \vec{r}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (\omega_2 z - \omega_3 y) & (\omega_3 x - \omega_1 z) & (\omega_1 y - \omega_2 x) \end{vmatrix}$$

Evaluating the determinant:

$$= 2(\omega_1 \vec{i} + \omega_2 \vec{j} + \omega_3 \vec{k}) = 2\vec{\omega}$$

Conclusion

Therefore:

$$\vec{\omega} = \frac{1}{2} \nabla \times \vec{v} = \frac{1}{2} \text{curl } \vec{v}$$

Physical Interpretation: The angular velocity of a rotating body is equal to one-half the curl of the linear velocity of any point of the body.

Chapter 2: Line Integrals and Green's Theorem in the Plane

Line Integrals

Setup

Let C be a curve in the (xy) -plane which connects points $A(a_1, b_1)$ and $B(a_2, b_2)$.

Let $P(x, y)$ and $Q(x, y)$ be single-valued functions defined at all points of C .

Subdivision of the Curve

Subdivide C into n parts by choosing $(n-1)$ points on it given by (x_1, y_1) , (x_2, y_2) , $(\dots)$, (x_{n-1}, y_{n-1}) .

Define: $\Delta x_k = x_k - x_{k-1}$ - $\Delta y_k = y_k - y_{k-1}$

for $(k = 1, 2, \dots, n)$, where: $(a_1, b_1) = (x_0, y_0)$ - $(a_2, b_2) = (x_n, y_n)$

Suppose that points (ξ_k, η_k) are chosen so that they are situated on C between points (x_{k-1}, y_{k-1}) and (x_k, y_k) .

Definition of Line Integral

Form the sum:

$$\left[\sum_{k=1}^n \left(P(x_k, y_k) \Delta x_k + Q(x_k, y_k) \Delta y_k \right) \right] \tag{2.1}$$

This sum approximates the line integral along curve (C) .

Definition of Line Integral

2D Line Integral

The limit of the sum as $(n \rightarrow \infty)$ in such a way that all the quantities (Δx_k) , (Δy_k) approach zero, is called a **line integral** along (C) and is denoted by:

$$\int_{(a_1, b_1)}^{(a_2, b_2)} P(x, y) \, dx + Q(x, y) \, dy$$

or

$$\int_{(a_1, b_1)}^{(a_2, b_2)} P \, dx + Q \, dy \tag{2.2}$$

3D Line Integral

A line integral along a curve (C) in three-dimensional space is defined as:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left[A_1(x_k, y_k, z_k) \Delta x_k + A_2(x_k, y_k, z_k) \Delta y_k + A_3(x_k, y_k, z_k) \Delta z_k \right]$$

$$= \int_C [A_1 \, dx + A_2 \, dy + A_3 \, dz] \tag{2.3}$$

where (A_1) , (A_2) and (A_3) are functions of (x) , (y) and (z) .

Vector Notation for Line Integrals

It is often convenient to express a line integral in vector form as an aid in physical or geometric understanding as well as for brevity of notation.

The 3D line integral can be written as:

$$\int_C [A_1 \, dx + A_2 \, dy + A_3 \, dz] = \int_C (A_1 \vec{i} + A_2 \vec{j} + A_3 \vec{k}) \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k})$$

$$= \int_C \vec{A} \cdot d\vec{r} \tag{2.4}$$

Parametric Form and Properties of Line Integrals

Parametric Form of Line Integral

If (C) is given in parametric form $(x = \phi(t), y = \psi(t))$, the line integral becomes:

$$\int_{t_1}^{t_2} \left[P(\phi(t), \psi(t)) \phi'(t) \, dt + Q(\phi(t), \psi(t)) \psi'(t) \, dt \right] \quad \text{tag{2.7}}$$

where (t_1) and (t_2) denote the values corresponding to points (A) and (B) respectively.

Properties of Line Integrals

Property 1: Separation

$$\int_C [P(x,y) \, dx + Q(x,y) \, dy] = \int_C P(x,y) \, dx + \int_C Q(x,y) \, dy$$

Property 2: Reversal of Path

$$\int_{(a_1, b_1)}^{(a_2, b_2)} [P \, dx + Q \, dy] = - \int_{(a_2, b_2)}^{(a_1, b_1)} [P \, dx + Q \, dy]$$

Property 3: Path Addition

$$\int_{(a_1, b_1)}^{(a_2, b_2)} [P \, dx + Q \, dy] = \int_{(a_1, b_1)}^{(a_3, b_3)} [P \, dx + Q \, dy] + \int_{(a_3, b_3)}^{(a_2, b_2)} [P \, dx + Q \, dy]$$

where $((a_3, b_3))$ is an intermediate point on the curve (C) .

Example 1: Line Integral Along Different Paths

Problem

Example (1): What is the value of $\int_A^B \frac{1}{x+y} \, dx$ along each of the paths shown in the figure?

The paths are: 1. Parabolic arc: $(y = x^2)$ from $(A(1,1))$ to $(B(2,4))$ 2. Piecewise linear path: from $(A(1,1))$ to $(Q(2,1))$ (horizontal), then from $(Q(2,1))$ to $(B(2,4))$ (vertical)

Solution

Before this integral can be evaluated, (y) must be expressed in terms of (x) . To do this, we recall from the definition of a line integral that the integrand is always to be evaluated **ALONG THE PATH OF INTEGRATION**.

Path 1: Parabolic Arc

Along the parabolic arc joining (A) and (B) , we have $(y = x^2)$.

The integral becomes:

$$\int_1^2 \frac{dx}{x + x^2} = \int_1^2 \frac{dx}{x(1 + x)}$$

Using partial fraction decomposition:

$$\frac{1}{x(1 + x)} = \frac{1}{x} - \frac{1}{1 + x}$$

Therefore:

$$\int_1^2 \left(\frac{1}{x} - \frac{1}{1 + x} \right) dx = \left[\ln x - \ln(1 + x) \right]_1^2$$

$$= [\ln 2 - \ln 3] - [\ln 1 - \ln 2]$$

$$= \ln 2 - \ln 3 + \ln 2 = \ln \left(\frac{4}{3} \right)$$

Answer for Path 1: $\boxed{\ln \left(\frac{4}{3} \right)}$

Example 1 (continued): Other Paths

Path 2: Straight Line from A to B

Along the straight-line path from (A) to (B) , we have $(y = 3x - 2)$, and making the substitution in the integrand, we obtain the ordinary definite integral:

$$\int_1^2 \frac{dx}{x + (3x - 2)} = \int_1^2 \frac{dx}{4x - 2}$$

$$= \frac{1}{4} \int_1^2 \frac{dx}{x - \frac{1}{2}} = \frac{1}{4} \left[\ln(4x - 2) \right]_1^2$$

$$= \frac{1}{4} [\ln(6) - \ln(2)] = \frac{1}{4} \ln 3$$

Answer for Path 2: $\boxed{\frac{1}{4} \ln 3}$

Path 3: Piecewise Linear Path APB

To compute the line integral along the path (APB) , we must perform two integrations, one along (AP) and one along (PB) , since the relation expressing (y) in terms of (x) is different on these two segments.

Along AP: The integral is zero, since (x) remains constant and therefore in the sum leading to the integral each (Δx_k) is zero.

Along PB: On which $(y = 4)$, we have the integral:

$$\int_1^2 \frac{dx}{x + 4} = \left[\ln(x + 4) \right]_1^2 = \ln 6 - \ln 5 = \ln \left(\frac{6}{5} \right)$$

Answer for Path 3: $\boxed{\ln \left(\frac{6}{5} \right)}$

Path 4: Piecewise Linear Path AQB

Along the path (AQB) we again have two integrations to perform.

Along AQ: On which $(y = 1)$, we have the integral:

$$\int_1^2 \frac{dx}{x + 1} = \left[\ln(x + 1) \right]_1^2 = \ln 3 - \ln 2 = \ln \left(\frac{3}{2} \right)$$

Along QB: The vertical segment $\overline{QB}$: the integral is again zero (since x is constant).

Answer for Path 4: $\boxed{\ln\left(\frac{3}{2}\right)}$

Example 2: Line Integral of Vector Field

Problem

Example (2): If $\vec{A} = (3x^2 - 6yz)\vec{i} + (2y + 3xz)\vec{j} + (1 - 4xyz^2)\vec{k}$, evaluate $\int_C \vec{A} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the following paths C :

- $x = t, y = t^2, z = t^3$
- The straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$, and then to $(1,1,1)$
- The straight line joining $(0,0,0)$ and $(1,1,1)$

Solution for Path (a)

The line integral is:

$$\int_C \vec{A} \cdot d\vec{r} = \int_C [(3x^2 - 6yz)\vec{i} + (2y + 3xz)\vec{j} + (1 - 4xyz^2)\vec{k}] \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$= \int_C [(3x^2 - 6yz) \, dx + (2y + 3xz) \, dy + (1 - 4xyz^2) \, dz]$$

Parameterization

If $x = t, y = t^2, z = t^3$, points $(0,0,0)$ and $(1,1,1)$ correspond to $t = 0$ and $t = 1$ respectively.

Then: $\vec{A} = (3t^2 - 6t^5)\vec{i} + (2t^2 + 3t^4)\vec{j} + (1 - 4t^9)\vec{k}$
 $\vec{r} = t\vec{i} + t^2\vec{j} + t^3\vec{k}$
 $d\vec{r} = (\vec{i} + 2t\vec{j} + 3t^2\vec{k}) \, dt$

Computing the Dot Product

$$\vec{A} \cdot d\vec{r} = (3t^2 - 6t^5)(1) + (2t^2 + 3t^4)(2t) + (1 - 4t^9)(3t^2)$$

$$= (3t^2 - 6t^5) + (4t^3 + 6t^5) + (3t^2 - 12t^{11})$$

$$= 6t^2 + 4t^3 - 12t^{11}$$

Evaluating the Integral

$$\int_C \vec{A} \cdot d\vec{r} = \int_0^1 (6t^2 + 4t^3 - 12t^{11}) \, dt$$

$$= \left[2t^3 + t^4 - t^{12} \right]_0^1 = 2 + 1 - 1 = 2$$

Answer for Path (a): $\boxed{2}$

Example 2 (continued): Path (b) - Piecewise Straight Lines

Path (b): Straight Lines from (0,0,0) to (0,0,1), then to (0,1,1), then to (1,1,1)

Segment 1: From (0,0,0) to (0,0,1)

Along the straight line from $((0,0,0))$ to $((0,0,1))$: - $(x = 0)$, $(y = 0)$, $(dx = 0)$, $(dy = 0)$ - (z) varies from (0) to (1)

The integral over this part of the path is:

$$\int_0^1 \left[3(0)^2 - 6(0)(z) \right] 0 + \left[2(0) + 3(0)(z) \right] 0 + \left[1 - 4(0)(0)(z^2) \right] dz$$

$$= \int_0^1 dz = 1$$

Segment 2: From (0,0,1) to (0,1,1)

Along the straight line from $((0,0,1))$ to $((0,1,1))$: - $(x = 0)$, $(z = 1)$, $(dx = 0)$, $(dz = 0)$ - (y) varies from (0) to (1)

The integral over this part of the path is:

$$\int_0^1 \left[3(0)^2 - 6(y)(1) \right] 0 + \left[2y + 3(0)(1) \right] dy + \left[1 - 4(0)(y)(1)^2 \right] 0$$

$$= \int_0^1 2y \, dy = \left[y^2 \right]_0^1 = 1$$

Segment 3: From (0,1,1) to (1,1,1)

Along the straight line from $((0,1,1))$ to $((1,1,1))$: - $(y = 1)$, $(z = 1)$, $(dy = 0)$, $(dz = 0)$ - (x) varies from (0) to (1)

The integral over this part of the path is:

$$\int_0^1 \left[(3x^2 - 6(1)(1))dx + (2(1) + 3x(1))0 + (1 - 4x(1)(1)^2)0 \right]$$

$$= \int_0^1 (3x^2 - 6) \, dx = \left[x^3 - 6x \right]_0^1 = 1 - 6 = -5$$

Total for Path (b)

Adding all three segments:

$$\int_C \vec{A} \cdot d\vec{r} = 1 + 1 - 5 = -3$$

Answer for Path (b): $\boxed{-3}$

Example 2 (continued): Path (c) - Straight Line

Path (c): Straight Line Joining (0,0,0) and (1,1,1)

The straight line joining $((0,0,0))$ and $((1,1,1))$ is given in parametric form by:

$$[x = t, \quad y = t, \quad z = t]$$

where (t) varies from (0) to (1) .

Then:

$$\int_C \vec{A} \cdot d\vec{r} = \int_0^1 \left[(3t^2 - 6t^2) \, dt + (2t + 3t^2) \, dt + (1 - 4t^4) \, dt \right]$$

Simplifying each term: - First term: $(3t^2 - 6t^2 = -3t^2)$ - Second term: $(2t + 3t^2)$ - Third term: $(1 - 4t^4)$

Combining:

$$= \int_0^1 (-3t^2 + 2t + 3t^2 + 1 - 4t^4) \, dt$$

$$\int_0^1 (2t + 1 - 4t^4) \, dt$$

$$\left[t^2 + t - \frac{4t^5}{5} \right]_0^1$$

$$= 1 + 1 - \frac{4}{5} = 2 - \frac{4}{5} = \frac{6}{5}$$

Answer for Path (c): $\boxed{\frac{6}{5}}$

Summary

- **Path (a):** $\int_C \vec{A} \cdot d\vec{r} = 2$
- **Path (b):** $\int_C \vec{A} \cdot d\vec{r} = -3$
- **Path (c):** $\int_C \vec{A} \cdot d\vec{r} = \frac{6}{5}$

Note: The line integral depends on the path taken, indicating that $\vec{A}$ is not a conservative vector field.

Simple Closed Curves and Connected Regions

Simple Closed Curves

A **simple closed curve** is a closed curve which does not intersect itself anywhere.

Examples: - A circle is simple - An irregular smooth closed loop (bean-shaped) is simple - A figure-eight shape is **not simple** (it intersects itself)

Simply and Multiply Connected Regions

If a plane region has the property that any closed curve in it can be continuously shrunk to a point without leaving the region, then the region is called **simply connected**, otherwise it is called **multiply connected**.

Examples of simply connected regions: - Interior of a circle - Exterior of a circle

Example of multiply connected region: - The area between two concentric circles

Positive Direction

The **positive direction** around a closed curve is defined as the direction in which an observer would move if he traversed the curve in such a way that the area of (R) was always on his left.

According to this definition: - If (R) is the interior of a simple closed curve (C) , then the positive direction around (C) is the **counter-clockwise** direction. - If (R) is the region exterior to a simple closed curve (C) , then the positive direction around (C) is the **clockwise** direction.

Green's Theorem in the Plane

Statement of Green's Theorem

Let (P) , (Q) , $(\frac{\partial P}{\partial y})$, $(\frac{\partial Q}{\partial x})$ be single-valued and continuous in a simply-connected region (R) bounded by a simple closed curve (C) . Then:

$$\oint_C [P \, dx + Q \, dy] = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy \tag{2.8}$$

where $\oint_C$ is used to emphasize that (C) is closed and that it is described in the positive direction.

Extension

This theorem is also true for regions bounded by two or more closed curves (i.e., multiply-connected regions).

Physical Interpretation

Green's Theorem relates a line integral around a closed curve to a double integral over the region enclosed by the curve. It is a special case of Stokes' Theorem in two dimensions.

Proof of Green's Theorem in the Plane

Setup

To prove Green's theorem in the plane, if (C) is a closed curve which has the property that any straight line parallel to the coordinate axes cuts (C) in at most two points.

Let the equations of the curves (AEB) and (AFB) be $(y = Y_1(x))$ and $(y = Y_2(x))$ respectively. If (R) is the region bounded by (C) , we have:

Proof for the $(\frac{\partial P}{\partial y})$ Term

$$\int_R \frac{\partial P}{\partial y} \, dx \, dy = \int_{x=a}^{x=b} \left[\int_{y=Y_1(x)}^{y=Y_2(x)} \frac{\partial P}{\partial y} \, dy \right] dx$$

Integrating with respect to (y) :

$$= \int_{x=a}^{x=b} \left[P(x, y) \right]_{y=Y_1(x)}^{y=Y_2(x)} dx$$

$$= \int_a^b [P(x, Y_2) - P(x, Y_1)] \, dx$$

Rearranging:

$$= \int_a^b P(x, Y_1) \, dx - \int_a^b P(x, Y_2) \, dx$$

Changing the limits on the second integral:

$$= \int_a^b P(x, Y_1) \, dx - \int_b^a P(x, Y_2) \, dx$$

Recognizing these as line integrals along (C_1) (lower curve) and (C_2) (upper curve):

$$= -\oint_C P \, dx$$

This proves the part of Green's Theorem involving $(\frac{\partial P}{\partial y})$.

A similar argument can be made for the $(\frac{\partial Q}{\partial x})$ term, leading to the complete statement of Green's Theorem.

Proof of Green's Theorem (continued)

Completion of the Proof

From the previous page, we had:

$$\oint_C P \, dx = - \iint_R \frac{\partial P}{\partial y} \, dx \, dy \tag{2.9}$$

Proof for the $\left(\frac{\partial Q}{\partial x}\right)$ Term

Similarly, let the equations of curves (EAF) and (EBF) be $(x = x_1(y))$ and $(x = x_2(y))$ respectively. Then:

$$\iint_R \frac{\partial Q}{\partial x} \, dx \, dy = \int_{y=e}^{y=f} \left[\int_{x=x_1(y)}^{x=x_2(y)} \frac{\partial Q}{\partial x} \, dx \right] dy$$

Integrating with respect to (x) :

$$= \int_e^f [Q(x_2(y), y) - Q(x_1(y), y)] \, dy$$

Rearranging:

$$= - \int_e^f Q(x_1(y), y) \, dy + \int_e^f Q(x_2(y), y) \, dy$$

Recognizing these as line integrals along the curve:

$$= \oint_C Q \, dy$$

Therefore:

$$\oint_C Q \, dy = \iint_R \frac{\partial Q}{\partial x} \, dx \, dy \tag{2.10}$$

Combining the Results

Adding equations (2.9) and (2.10):

$$\oint_C [P \, dx + Q \, dy] = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy$$

This completes the proof of Green's Theorem.

Extension of Green's Theorem to Complex Curves

Problem

To extend the proof to curves for which lines parallel to the coordinate axes may cut the curve in more than two points.

Method

Consider a closed curve (C) such as shown in the figure. By constructing line (ST) , the region is divided into two regions which are of the type considered in the previous proof and for which Green's theorem applies, i.e.:

$$\int_{STUS} [P \, dx + Q \, dy] = \iint_{R_1} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy \tag{2.11}$$

$$\int_{SVTS} [P \, dx + Q \, dy] = \iint_{R_2} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy \tag{2.12}$$

Adding the Left-Hand Sides

Adding the left-hand sides:

$$\int_{STUS} + \int_{SVTS} = \int_{ST} + \int_{TUS} + \int_{SVT} + \int_{TS}$$

Since $(\int_{ST} + \int_{TS} = 0)$ (the integrals cancel as they are in opposite directions):

$$\int_{TUS} + \int_{SVT} = \oint_C$$

Adding the Right-Hand Sides

Adding the right-hand sides:

$$\iint_{R_1} + \iint_{R_2} = \iint_R$$

Conclusion

Then:

$$\oint_C [P \, dx + Q \, dy] = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy$$

This extends Green's Theorem to more complex curves.

Green's Theorem for Multiply-Connected Regions

Setup

To extend the theorem to multiply-connected regions, consider the shaded region shown in the figure.

The region is multiply-connected since not every closed curve can be shrunk to a point without leaving (R) , as is observed by considering a curve surrounding $(DEFGD)$ for example.

Boundary and Positive Direction

The boundary of (R) , which consists of the exterior boundary $(AHJKLA)$ and the interior boundary $(DEFGD)$, is to be traversed in the positive direction, so that a person traveling in this direction always has the region on his left. It is seen that the positive directions are those indicated in the figure.

Construction of Cross-Cut

In order to establish the theorem, construct a line, such as (AD) , called a **cross-cut**, connecting the exterior and interior boundaries. The region bounded by $(ADEF GDALKJHA)$ is simply-connected, and so Green's theorem is valid.

Application of Green's Theorem

Then:

$$\oint_{ADEF GDALKJHA} [P \, dx + Q \, dy] = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy$$

But the integral on the left (leaving out the integrand) is equal to:

$$\int_{AD} + \int_{DEFGD} + \int_{DA} + \int_{ALKJHA} = \int_{DEFGD} + \int_{ALKJHA}$$

since $(\int_{AD} + \int_{DA} = 0)$ (the integrals cancel as they are in opposite directions).

Final Result

Thus, if (C_1) is the curve $(ALKJHA)$, (C_2) is the curve $(DEFGD)$, and (C) is the boundary of (R) consisting of (C_1) and (C_2) (traversed in the positive directions), then:

$$\oint_{C_1} + \oint_{C_2} = \oint_C$$

and so:

$$\oint_C [P \, dx + Q \, dy] = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy$$

Green's Theorem in Vector Notation

Vector Form of the Line Integral

To express Green's theorem in the plane in vector notation:

We have:

$$[P \, dx + Q \, dy = (P\vec{i} + Q\vec{j}) \cdot (dx\vec{i} + dy\vec{j}) = \vec{A} \cdot d\vec{r}]$$

where: $(\vec{A} = P\vec{i} + Q\vec{j}) - (\vec{r} = x\vec{i} + y\vec{j})$, so $(d\vec{r} = dx\vec{i} + dy\vec{j})$

Curl of A in 2D

Also:

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & 0 \end{vmatrix}$$

Expanding the determinant:

$$= \left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(Q) \right) \vec{i} - \left(\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(P) \right) \vec{j} + \left(\frac{\partial}{\partial x}(Q) - \frac{\partial}{\partial y}(P) \right) \vec{k}$$

Since (P) and (Q) are functions of (x) and (y) only:

$$= \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$$

Dot Product with Unit Vector

$$((\nabla \times \vec{A}) \cdot \vec{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})$$

Green's Theorem in Vector Form

Then Green's theorem in the plane can be written:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_R (\nabla \times \vec{A}) \cdot \vec{k} \, dx \, dy \quad \text{tag{2.13}}$$

Area Calculation Using Green's Theorem

Area Formula

The area of a region (R) bounded by a simple closed curve (C) can be expressed using Green's Theorem.

If we choose $(P = 0)$ and $(Q = x)$, then:

$$\oint_C x \, dy = \iint_R \left(\frac{\partial x}{\partial x} - \frac{\partial 0}{\partial y} \right) dx \, dy = \iint_R dx \, dy = \text{Area of } R$$

Alternatively, if we choose $(P = -y)$ and $(Q = 0)$, then:

$$\oint_C (-y) \, dx = \iint_R \left(\frac{\partial 0}{\partial x} - \frac{\partial (-y)}{\partial y} \right) dx \, dy = \iint_R dx \, dy = \text{Area of } R$$

Combined Formula

Combining both formulas:

$$\text{Area} = \frac{1}{2} \oint_C (x \, dy - y \, dx)$$

This provides a convenient way to calculate the area of a region using a line integral around its boundary.

Area Formula and Example: Verifying Green's Theorem

Area Formula

The area bounded by a simple closed curve (C) is given by:

$$A = \frac{1}{2} \oint_C [x \, dy - y \, dx]$$

Derivation:

Putting $(P = -y)$, $(Q = x)$, then:

$$\oint_C [x \, dy - y \, dx] = \iint_R \left[\frac{\partial}{\partial x} (x) - \frac{\partial}{\partial y} (-y) \right] dx \, dy$$

$$[= \iint_R [1 - (-1)] dx \, dy = 2 \iint_R dx \, dy = 2A]$$

where (A) is the required area.

Thus:

$$A = \frac{1}{2} \oint_C [x \, dy - y \, dx]$$

Example: Verify Green's Theorem

Verify Green's theorem in the plane for:

$$\oint_C [(2xy - x^2) \, dx + (x + y^2) \, dy]$$

where (C) is the closed curve of the region bounded by $(y = x^2)$ and $(y^2 = x)$.

Intersection Points: The plane curves $(y = x^2)$ and $(y^2 = x)$ intersect at $((0,0))$ and $((1,1))$.

Positive Direction: The positive direction in traversing (C) is as shown in the figure (counter-clockwise).

Example: Verification of Green's Theorem (Solution)

Solution

Method 1: Direct Line Integral Calculation

Along $(y = x^2)$ (from $((0,0))$ to $((1,1))$):

The line integral equals:

$$\int_{x=0}^1 \left[2x(x^2) - x^2 \right] dx + \int_0^1 (x + (x^2)^2) d(x^2)$$

$$= \int_0^1 \left[(2x^3 - x^2) dx + (x + x^4)(2x \, dx) \right]$$

$$= \int_0^1 (2x^3 - x^2 + 2x^2 + 2x^5) dx = \int_0^1 (2x^3 + x^2 + 2x^5) dx$$

$$= \left[\frac{x^4}{2} + \frac{x^3}{3} + \frac{x^6}{3} \right]_0^1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{3} = \frac{7}{6}$$

Along $(y^2 = x)$ (from $((1,1))$ to $((0,0))$):

The line integral equals:

$$\int_{y=1}^0 \left[2(y^2)(y) - (y^2)^2 \right] d(y^2) + \int_1^0 (y^2 + y^2) dy$$

$$= \int_1^0 \left[(2y^3 - y^4)(2y \, dy) + 2y^2 dy \right]$$

$$= \int_1^0 (4y^4 - 2y^5 + 2y^2) dy = - \int_0^1 (4y^4 - 2y^5 + 2y^2) dy$$

$$= - \left[\frac{4y^5}{5} - \frac{2y^6}{6} + \frac{2y^3}{3} \right]_0^1 = - \left(\frac{4}{5} - \frac{1}{3} + \frac{2}{3} \right) = - \frac{17}{15}$$

Total Line Integral:

$$\oint_C = \frac{7}{6} - \frac{17}{15} = \frac{35 - 34}{30} = \frac{1}{30}$$

Method 2: Using Green's Theorem (Double Integral)

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy = \iint_R \left(\frac{\partial}{\partial x} (x + y^2) - \frac{\partial}{\partial y} (2xy - x^2) \right) dx \, dy$$

$$= \iint_R (1 - 2x) dx \, dy$$

Setting up the iterated integral:

$$= \int_{x=0}^1 \int_{y=x^2}^{\sqrt{x}} (1 - 2x) dy \, dx$$

$$= \int_0^1 [y - 2xy]_{y=x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 [(\sqrt{x} - 2x^{3/2}) - (x^2 - 2x^3)] dx$$

$$= \int_0^1 (x^{1/2} - 2x^{3/2} - x^2 + 2x^3) dx$$

$$= \left[\frac{2x^{3/2}}{3} - \frac{4x^{5/2}}{5} - \frac{x^3}{3} + \frac{x^4}{2} \right]_0^1$$

$$= \frac{2}{3} - \frac{4}{5} - \frac{1}{3} + \frac{1}{2} = \frac{1}{30}$$

Conclusion

Both methods yield the same result: $\frac{1}{30}$. **Green's theorem is verified.**

Independence of the Path for a Line Integral

Necessary and Sufficient Condition

A necessary and sufficient condition for $\int [P \, dx + Q \, dy]$ to be independent of the path (C) joining any two given points in a region (R) is that in (R) :

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \text{tag{2.14}}$$

where it is supposed that these partial derivatives are continuous in (R) .

Exact Differential

The condition (2.14) is also the condition that $(P \, dx + Q \, dy)$ is an **exact differential**, i.e., that there exists a function $\phi(x, y)$ such that:

$$P \, dx + Q \, dy = d\phi$$

Evaluation of the Line Integral

In such case, if the end points of curve (C) are (x_1, y_1) and (x_2, y_2) , the value of the line integral is given by:

$$\int_{(x_1, y_1)}^{(x_2, y_2)} (P \, dx + Q \, dy) = \int_{(x_1, y_1)}^{(x_2, y_2)} d\phi = \phi(x_2, y_2) - \phi(x_1, y_1) \tag{2.15}$$

Special Case: Closed Path

In particular, if (2.14) holds and (C) is closed, we have:

$$\oint_C (P \, dx + Q \, dy) = 0 \tag{2.16}$$

Proof: Path Independence Condition

Sufficiency

Suppose $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. Then by Green's theorem:

$$\oint_C (P \, dx + Q \, dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy = 0$$

where (R) is the region bounded by (C) .

Since the integrand is zero, the line integral around any closed path is zero, which implies path independence.

Necessity

Suppose $\oint_C [P \, dx + Q \, dy] = 0$ around every closed path (C) in (R) and that $\left(\frac{\partial P}{\partial y}\right)$ and $\left(\frac{\partial Q}{\partial x}\right)$ are continuous in (R) .

In particular, suppose $\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} > 0\right)$ at the point $((x_0, y_0))$.

By hypothesis, $\left(\frac{\partial P}{\partial y}\right)$ and $\left(\frac{\partial Q}{\partial x}\right)$ are continuous in (R) , so that there must be some region (τ) containing $((x_0, y_0))$ as an interior point for which $\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} > 0\right)$.

If (Γ) is the boundary of (τ) , then by Green's theorem:

$$\oint_{\Gamma} [P \, dx + Q \, dy] = \iint_{\tau} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \, dy > 0$$

which contradicts the hypothesis that $\oint_C [P \, dx + Q \, dy] = 0$ for **all** closed curves in (R) .

Conclusion

Thus $\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$ cannot be positive. Similarly, it cannot be negative, and it follows that it must be identically zero, i.e.:

$$\left(\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}\right)$$

identically in (R) .

Completion of the Path-Independence Proof

From the previous arguments we saw that if, in a region (R) ,

$$\oint_C [P \, dx + Q \, dy] = 0$$

for every closed curve (C) in (R) , then

$$\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0\right)$$

cannot be **positive** anywhere in (R) .

Similarly, it cannot be **negative** anywhere in (R) , and therefore it must be identically zero; that is,

$$\left[\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \right]$$

identically in (R) .

To prove that this condition is also **sufficient** for

$$\left[\int_A^B [P \, dx + Q \, dy] \right]$$

to be independent of the path in (R) joining the points (A) and (B) , assume that

$$\left[\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \right]$$

everywhere in (R) .

From the result just obtained, for any closed curve (C) in (R) we then have

$$\left[\oint_C [P \, dx + Q \, dy] = 0 . \right]$$

Let (C_1) and (C_2) be two different paths from (A) to (B) contained in (R) , and let (C) be the closed curve obtained by traversing (C_1) from (A) to (B) and then (C_2) from (B) back to (A) .

Along this closed curve

$$\left[\oint_C [P \, dx + Q \, dy] = \int_{C_1} [P \, dx + Q \, dy] + \int_{C_2} [P \, dx + Q \, dy] = 0 . \right]$$

Hence

$$\left[\int_{C_1} [P \, dx + Q \, dy] = - \int_{C_2} [P \, dx + Q \, dy], \right]$$

but the second integral is taken from (B) to (A) . Reversing its direction gives

$$\left[\int_A^B {}_{C_1} [P \, dx + Q \, dy] = \int_A^B {}_{C_2} [P \, dx + Q \, dy], \right]$$

and therefore the integral is **independent of the path** joining (A) and (B) in (R) .

Example: Path-Independent Line Integral

Prove that

Example (continued): Method (2)

Finding a Potential Function

Since

$$\left[\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \right]$$

there exists a scalar potential function $\phi(x,y)$ such that

$$[P \, dx + Q \, dy = d\phi .]$$

From

$$[P = \frac{\partial \phi}{\partial x} = 6xy^2 - y^3]$$

we integrate with respect to x (treating y as constant):

$$[\phi(x,y) = 3x^2y^2 - xy^3 + f(y),]$$

where $f(y)$ is a function of y only.

Similarly, from

$$[Q = \frac{\partial \phi}{\partial y} = 6x^2y - 3xy^2,]$$

integration with respect to y gives

$$[\phi(x,y) = 3x^2y^2 - x y^3 + g(x),]$$

where $g(x)$ is a function of x only.

The only way these two expressions for ϕ can be equal for all (x,y) is that

$$[f(y) = g(x) = c,]$$

a constant. Hence we may take

$$[\phi(x,y) = 3x^2y^2 - xy^3 + c .]$$

Evaluating the Line Integral

Then

$$[\int_{(1,2)}^{(3,4)} [(6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy] = \phi(3,4) - \phi(1,2).]$$

That is,

$$\phi(3,4) = 3 \cdot 3^2 \cdot 4^2 - 3 \cdot 4^3 + c = 432 - 192 + c = 240 + c,$$

$$\phi(1,2) = 3 \cdot 1^2 \cdot 2^2 - 1 \cdot 2^3 + c = 12 - 8 + c = 4 + c.$$

Hence

$$\phi(3,4) - \phi(1,2) = (240 + c) - (4 + c) = 236.$$

Thus, by Method (2), the value of the line integral is again

$$\boxed{236}.$$

Complex Variables and Conformal Mapping

Complex Numbers

A **complex number** is a number of the form

$$z = a + bi,$$

where a and b are real numbers and i , called the **imaginary unit**, has the property that $i^2 = -1$.

If $z = a + bi$, then:

- a is called the **real part** of z ,
- b is called the **imaginary part** of z .

The symbol z , which represents any complex number, is called a **complex variable**.

Two complex numbers $a + bi$ and $c + di$ are **equal** if and only if

$$a = c \quad \text{and} \quad b = d.$$

The **complex conjugate** (or simply *conjugate*) of a complex number $a + bi$ is

$$\overline{z} = a - bi.$$

The conjugate of a complex number z is often denoted by $\overline{z}$.

Fundamental Operations with Complex Numbers

1. Addition

For $(z_1 = a + bi)$ and $(z_2 = c + di)$,

$$\backslash [z_1 + z_2 = (a + bi) + (c + di) = (a + c) + (b + d)i . \backslash$$

1. Subtraction

$$\backslash (a + bi) - (c + di) = (a - c) + (b - d)i . \backslash$$

Further operations (multiplication, division, modulus, argument, etc.) continue on the next pages.

Fundamental Operations with Complex Numbers (continued)

1. Multiplication

Let

$$[z_1 = a + bi, \quad z_2 = c + di.]$$

Then

$$\backslash \text{ z_1 z_2} = (a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i . \backslash$$

1. Division

For $(z_1 = a + bi)$ and $(z_2 = c + di \neq 0)$,

$$\begin{aligned} \left[\frac{a+bi}{c+di} \right] &= \frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac-bd+bc i - ad i^2}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2} i. \end{aligned}$$

Examples

1. $\lfloor (7 - i) + (3 + 2i) = 7 + 3 + (-1 + 2)i = 10 + i. \rfloor$

$$2. \quad [(5 + 3i) + \big[(1 + 2i) + (7 - 5i)\big]] = (5 + 3i) + (8 - 3i) = 13.$$

3. $\sqrt{(2 - 3i)(4 + 2i)} = 2(4 + 2i) - 3i(4 + 2i) = 8 + 4i - 12i - 6i^2 = 14 - 8i.$

$$4. \left[\frac{3-2i}{-1+i} = \frac{3-2i}{-1+i} \cdot \frac{-1-i}{-1-i} = \frac{-3-3i+2i+2i^2}{1-i^2} = \frac{-5-i}{2} = -\frac{5}{2} - \frac{1}{2}i \right]$$

Geometric Representation of Complex Numbers

A complex number

$$[z = x + iy]$$

can be represented as a point (P) with coordinates $((x,y))$ in a rectangular (xy) -plane called the **complex plane** or **Argand diagram**.

The number (z) can also be represented as a vector $(\overrightarrow{OP})$ which joins the origin (O) to (P) .

Let

$$[x = r\cos\theta, \quad y = r\sin\theta,]$$

where

$$[r = \sqrt{x^2 + y^2}]$$

is the **modulus** (or absolute value) of (z) , and

$$[\theta = \tan^{-1}\frac{y}{x}]$$

is the **argument** (or amplitude) of (z) .

Since $(z = x + iy)$, we may write

$$[z = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta) = r\operatorname{cis}\theta .]$$

This is known as the **polar form** of a complex number.

Geometric Interpretation of Basic Operations

Let $(z = a + ib)$ be represented by the point $((a,b))$ in the complex plane.

- The point representing the **conjugate** $(\overline{z} = a - ib)$ is the reflection of $((a,b))$ in the real axis.
- The point representing $(-z = -a - ib)$ is the reflection of $((a,b))$ in the origin.

If z_1 and z_2 are represented by vectors $\vec{Oz_1}$ and $\vec{Oz_2}$, then:

- The vector for $z_1 + z_2$ is the diagonal of the parallelogram constructed on $\vec{Oz_1}$ and $\vec{Oz_2}$.
- The vector for $z_1 - z_2$ is obtained by adding z_1 and $-z_2$, i.e. by the parallelogram rule with $\vec{O(-z_2)}$.

These diagrams illustrate addition and subtraction of complex numbers as vector addition and subtraction in the Argand plane.

Example: Rectangular to Polar Form

Example. Express each of the following complex numbers in polar form.

(a) $(2 + 2i)$

Modulus

$$|r| = |2 + 2\sqrt{3}i| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4.$$

Argument

$$|\theta = \sin^{-1}\frac{2\sqrt{3}}{4} = \sin^{-1}\frac{\sqrt{3}}{2} = 60^\circ = \frac{\pi}{3}.$$

Hence

$$2 + 2\sqrt{3}i = r(\cos\theta + i\sin\theta) = 4\big(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\big).$$

(b) $(-\sqrt{6} - \sqrt{2}i)$

Modulus

$$|r| = |-\sqrt{6} - \sqrt{2}i| = \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} = \sqrt{6 + 2} = 2\sqrt{2}.$$

The point lies in the third quadrant. The reference angle is (30°) , so

$$|\theta = 180^\circ + 30^\circ = 210^\circ = \frac{7\pi}{6}.$$

Thus

$$\left[-\sqrt{6} - \sqrt{2}, i = 2\sqrt{2} \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) \right]$$

Absolute Value (Modulus)

The **absolute value** or **modulus** of a complex number $(a + bi)$ is defined as

$$\left[|a + bi| = \sqrt{a^2 + b^2} \right]$$

If (z_1) and (z_2) are complex numbers, the following properties hold:

$$1. \left[|z_1 z_2| = |z_1| |z_2| \right]$$

$$2. \left[\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0) \right]$$

3. Triangle inequality

$$\left[|z_1 + z_2| \leq |z_1| + |z_2| \right]$$

$$1. \left[|z_1 - z_2| \geq \big| |z_1| - |z_2| \big| \right]$$

Euler's Formula

The Taylor series expansion for (e^x) is

$$\left[e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \right]$$

Substituting $(x = i\theta)$ gives

$$\left[e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \cdots \right]$$

which separates into real and imaginary parts:

$$\left[e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots \right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots \right) = \cos \theta + i \sin \theta \right]$$

This fundamental relation

$$\left[e^{i\theta} = \cos \theta + i \sin \theta \right]$$

is called **Euler's formula**.

De Moivre's Theorem

Let

$$[z_1 = x_1 + iy_1 = r_1(\cos\theta_1 + i\sin\theta_1),] [z_2 = x_2 + iy_2 = r_2(\cos\theta_2 + i\sin\theta_2).]$$

Then

$$[z_1 z_2 = r_1(\cos\theta_1 + i\sin\theta_1), r_2(\cos\theta_2 + i\sin\theta_2)] \\ [= r_1 r_2 \{ \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2 + i(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2) \}] [= r_1 r_2 \{ \cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2) \}.]$$

If $(z_1 = z_2 = \dots = z_n = z = r(\cos\theta + i\sin\theta))$, then by repeated multiplication,

$$[z^n = \{ r(\cos\theta + i\sin\theta) \}^n = r^n \big(\cos n\theta + i\sin n\theta \big).]$$

This is known as **De Moivre's Theorem**.

Example

Find all values of (z) for which

$$[z^5 = -32,]$$

and locate these values in the complex plane.

In polar form

$$[-32 = 32 \{ \cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi) \}, \quad k = 0, \pm 1, \pm 2, \dots]$$

Continuing the example $(z^5 = -32)$.

Let

$$[z = r(\cos\theta + i\sin\theta).]$$

Then by De Moivre's Theorem,

$$[z^5 = r^5(\cos 5\theta + i\sin 5\theta) = -32 = 32 \{ \cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi) \}.]$$

Thus

$$r^5 = 32, \quad r = 2,$$

and

$$\theta = \pi + 2k\pi, \quad \theta = \frac{\pi + 2k\pi}{5}, \quad k = 0, 1, 2, 3, 4.$$

Hence the five distinct roots are

$$z_k = 2 \left(\cos \frac{\pi + 2k\pi}{5} + i \sin \frac{\pi + 2k\pi}{5} \right), \quad k = 0, 1, 2, 3, 4.$$

Explicitly,

$$\begin{aligned} z_1 &= 2 \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right), \quad z_2 = 2 \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right), \quad z_3 = 2 \left(\cos \frac{5\pi}{5} + i \sin \frac{5\pi}{5} \right) = 2(\cos \pi + i \sin \pi), \\ z_4 &= 2 \left(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} \right), \quad z_5 = 2 \left(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} \right). \end{aligned}$$

By considering $(k = 5, 6, \dots)$ as well as negative values $(k = -1, -2, \dots)$, repetitions of these five values are obtained.

Hence these are the **only solutions** (or **roots**) of the given equation.

They are called the **fifth roots of (-32)** and are equally spaced on a circle of radius (2) in the Argand plane.

Variables and Functions

A symbol such as (z) , which can stand for any one of a set of complex numbers, is called a **complex variable**.

If to each value which a complex variable (z) can assume there corresponds one or more values of a complex variable (w) , we say that **(w) is a function of (z)** and write

$$w = f(z) \quad \text{or} \quad w = g(z), \quad \text{etc.}$$

A function is called **single-valued** if for each value of (z) there corresponds only one value of (w) ; otherwise it is **multiple-valued**.

Unless otherwise specified we shall assume that $(f(z))$ is single-valued.

In general we can write

$$[w = f(z) = u(x,y) + iv(x,y),]$$

where (u) and (v) are real functions of (x) and (y) .

Example (1)

Let

$$[w = z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy.]$$

Then

$$[u(x,y) = x^2 - y^2, \quad v(x,y) = 2xy.]$$

These are called the **real** and **imaginary** parts of $(w = z^2)$, respectively.

Example (2)

Let

$$[w = \frac{1}{1 - z} = \frac{1}{1 - (x + iy)} = \frac{1}{1 - x - iy}.]$$

Multiply numerator and denominator by the conjugate $(1 - x + iy)$:

$$[w = \frac{1 - x + iy}{(1 - x)^2 + y^2}.]$$

Hence

$$[u(x,y) = \frac{1 - x}{(1 - x)^2 + y^2}, \quad v(x,y) = \frac{y}{(1 - x)^2 + y^2}.]$$

Limits

The function $(f(z))$ is said to have the **limit** (L) as (z) approaches (z_0) , and we write

$$[\lim_{z \rightarrow z_0} f(z) = L,]$$

if for any positive number (ϵ) (however small) we can find some positive number (δ) such that

$$[|f(z) - L| < \epsilon]$$

whenever

$$[0 < |z - z_0| < \delta .]$$

Continuity

The function $f(z)$ is said to be **continuous** at $(z = z_0)$ if

$$[\lim_{z \rightarrow z_0} f(z) = f(z_0).]$$

This implies three conditions in order that $f(z)$ be continuous at $(z = z_0)$:

1. $\lim_{z \rightarrow z_0} f(z) = L$ must exist.
2. $f(z_0)$ must exist, i.e. $f(z)$ is defined at (z_0) .
3. $(L = f(z_0))$.

Derivatives

If $f(z)$ is single-valued in some region (R) of the (z) -plane, the **derivative** of $f(z)$ is defined as

$$[f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}, \tag{1}]$$

provided that this limit exists **independently of the manner in which** $(\Delta z \rightarrow 0)$.

Analytic Functions

If the derivative $f'(z)$ exists at all points of a region (R) , then $f(z)$ is said to be **analytic in (R)** .

Cauchy–Riemann Equations

A necessary condition that

$$[w = f(z) = u(x,y) + iv(x,y)]$$

be analytic in a region (R) is that, in (R) , (u) and (v) satisfy the **Cauchy–Riemann equations**

$$[\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{2}]$$

If the second partial derivatives of u and v with respect to x and y exist and are continuous, we find by differentiating (2) that

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0, \\ \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0. \end{aligned}$$

Thus the real and imaginary parts satisfy **Laplace's equation** in two dimensions.

Functions satisfying Laplace's equation are called **harmonic functions**.

Proof of the Cauchy–Riemann Equations (Sketch)

Since

$$f(z) = f(x + iy) = u(x, y) + i v(x, y),$$

we have

$$f(z + \Delta z) = f(x + \Delta x + i(y + \Delta y)) = u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y),$$

where $\Delta z = \Delta x + i \Delta y$.

Then, from the definition of the derivative,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\substack{\Delta x \rightarrow 0, \\ \Delta y \rightarrow 0}} \frac{[u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta x + i \Delta y}.$$

If $\Delta y = 0$ (approach along the x -direction), the required limit is

$$\lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

If $\Delta x = 0$ (approach along the y -direction), the required limit is

$$\lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} = \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} = -i \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}.$$

If the derivative $(f'(z))$ is to exist, these two special limits must be equal; that is,

$$\left[\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}, \right]$$

which yields the Cauchy–Riemann equations

$$\left[\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \right]$$

Examples Using the Cauchy–Riemann Equations

Example (4)

For

$$\left[f(z) = \overline{z} = x - iy, \right]$$

we have

$$\left[u = x, \quad v = -y. \right]$$

Then

$$\left[\frac{\partial u}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = -1. \right]$$

Only the second of the Cauchy–Riemann equations,

$$\left[\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \right]$$

is satisfied; the first is not. Hence there is **no point** in the (z) -plane where $(f'(z))$ exists, and therefore $(f(z) = \overline{z})$ is nowhere analytic.

Example (5)

For

$$\left[f(z) = z\overline{z} = x^2 + y^2, \right]$$

we have

$$\left[u = x^2 + y^2, \quad v = 0. \right]$$

Thus

$$\left[\begin{aligned} \frac{\partial u}{\partial x} &= 2x, & \frac{\partial u}{\partial y} &= 2y, \\ \frac{\partial v}{\partial x} &= 0, & \frac{\partial v}{\partial y} &= 0. \end{aligned} \right]$$

The Cauchy–Riemann equations become

$$\left[\begin{aligned} 2x &= 0, & 2y &= 0, \end{aligned} \right]$$

which are satisfied **only at the origin**. Hence $(z = 0)$ is the only point at which $(f'(z))$ exists, and therefore $(f(z) = \overline{z})$ is nowhere analytic.

Applying the Cauchy–Riemann Equations

For functions which are known to have continuous first partial derivatives, the Cauchy–Riemann equations provide a convenient test for analyticity:

1. Express $(f(z))$ in the form $\left[f(z) = u(x,y) + iv(x,y), \right]$ where $(z = x + iy)$.
2. Compute the first partial derivatives $\left[\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \right]$.
3. Determine the points $((x,y))$ at which the Cauchy–Riemann equations $\left[\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \right]$ are satisfied.

If, in addition, the partial derivatives are continuous, then $(f(z))$ is **analytic** at all such points and nowhere else.

In subsequent sections these equations will be used to construct analytic functions, to relate potential and stream functions in fluid flow, and to solve boundary–value problems in two dimensions.

Example (6)

For

$$\left[f(z) = z^2 = (x^2 - y^2) + 2ixy, \right]$$

we have

$$\left[u = x^2 - y^2, \quad v = 2xy, \right]$$

and therefore

$$\left[\begin{aligned} \frac{\partial u}{\partial x} &= 2x, & \frac{\partial u}{\partial y} &= -2y, \\ \frac{\partial v}{\partial x} &= 2y, & \frac{\partial v}{\partial y} &= 2x. \end{aligned} \right]$$

The Cauchy–Riemann equations are identically satisfied.

Hence the derivative $(f'(z))$ exists at all points of the (z) –plane, and its value is

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \\ &= \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} = 2x + 2iy = 2z. \end{aligned}$$

This is exactly what formal differentiation according to the power rule would give.

Example (7)

Prove that in polar form the Cauchy–Riemann equations can be written

$$\left[\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{r} \frac{\partial v}{\partial \theta}, \\ \frac{\partial v}{\partial r} &= -\frac{1}{r} \frac{\partial u}{\partial \theta}. \end{aligned} \right]$$

We use the relations

$$\left[\begin{aligned} x &= r \cos \theta, \\ y &= r \sin \theta, \end{aligned} \right]$$

so that

$$\left[\begin{aligned} r^2 &= x^2 + y^2, \\ \tan \theta &= \frac{y}{x}. \end{aligned} \right]$$

By the chain rule we can express

$$\left[\begin{aligned} \frac{\partial u}{\partial x}, & \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \\ \frac{\partial v}{\partial y} \end{aligned} \right]$$

in terms of $(\frac{\partial u}{\partial r})$, $(\frac{\partial u}{\partial \theta})$ and $(\frac{\partial v}{\partial r})$, $(\frac{\partial v}{\partial \theta})$.

Substituting these expressions into the Cauchy–Riemann equations and solving the resulting simultaneous equations yields the polar form displayed above.

Conformal Mapping

Transformations or Mappings

The set of equations

$$\begin{cases} u = u(x,y), \\ v = v(x,y) \end{cases} \quad (4)$$

defines in general a **transformation** or **mapping** which establishes a correspondence between points in the (u,v) - and (x,y) -planes.

The equations (4) are called the **transformation equations**.

If to each point of the (u,v) -plane there corresponds one and only one point of the (x,y) -plane, we speak of a **one-to-one** transformation or mapping.

In such a case a set of points in the (x,y) -plane is “wrapped” into a set of points in the (u,v) -plane and conversely.

The corresponding sets of points in the two planes are often called **images** of each other.

Conformal Mapping (continued)

Suppose that under a transformation (4) a point $((x_0, y_0))$ of the (x,y) -plane is mapped into a point $((u_0, v_0))$ of the (u,v) -plane, while curves (C_1) and (C_2) (intersecting at $((x_0, y_0))$) are mapped respectively into curves (C_1') and (C_2') (intersecting at $((u_0, v_0))$).

If the transformation is such that the angle at $((x_0, y_0))$ between (C_1) and (C_2) is equal to the angle at $((u_0, v_0))$ between (C_1') and (C_2') , both in magnitude and in sense, the transformation or mapping is said to be **conformal** at $((x_0, y_0))$.

The following theorem is fundamental.

Theorem

If $(f(z))$ is analytic and $(f'(z) \neq 0)$ in a region (R) , then the mapping

$$w = f(z)$$

is **conformal** at all points of (R) .

Riemann's Mapping Theorem

Let (C) be a simple closed curve in the (z) -plane forming the boundary of a region (R) .

Let (C') be a circle of radius one and centre at the origin (the **unit circle**) forming the boundary of a region (R') in the (w) -plane. The region (R') is sometimes called the **unit disk**.

Then **Riemann's mapping theorem** states that there exists a function

$$[w = f(z),]$$

analytic in (R) , which maps each point of (R) into a corresponding point of (R') and each point of (C) into a corresponding point of (C') , the correspondence being one-to-one.

This function $(f(z))$ contains three arbitrary real constants which can be determined by making the centre of (C') correspond to some given point in (R) , while some point on (C') corresponds to a given point on (C) .

It should be noted that while Riemann's mapping theorem demonstrates the **existence** of such a mapping, it does not in general provide an explicit formula for $(f(z))$.

Some General Transformations

In the following, (α) and (β) are given complex constants, while (a) and (θ) are real constants.

1. Translation

$$[w = z + \beta.]$$

By this transformation, figures in the (z) -plane are **displaced** or **translated** in the direction of the vector (β) .

2. Rotation

$$[w = e^{i\theta} z.]$$

By this transformation, figures in the (z) -plane are rotated through an angle (θ) .

If $(\theta > 0)$ the rotation is counter-clockwise; if $(\theta < 0)$ the rotation is clockwise.

3. Stretching

$$[w = az.]$$

By this transformation, figures in the (z) -plane are stretched (or contracted) in the direction of (z) if $(a > 1)$ (or $(0 < a < 1)$).

We consider contraction as a special case of stretching.

4. Inversion

$$[w = \frac{1}{z}.]$$

5. Linear Transformation

$$[w = \alpha z + \beta.]$$

This is a combination of the transformations of translation, rotation and stretching.

1. Bilinear or Fractional Transformation

$$[w = \frac{\alpha z + \beta}{\gamma z + \delta}, \quad \alpha\delta - \beta\gamma \neq 0.]$$

This is a combination of the transformations of translation, rotation, stretching and inversion.

Example (1)

Let the rectangular region (R) in the (z) -plane be bounded by

$$[x = 0, \quad y = 0, \quad x = 2, \quad y = 1.]$$

Determine the region (R') of the (w) -plane into which (R) is mapped under the transformations:

1. $(w = z + (1 + 2i))$
2. $(w = \sqrt{2}e^{i\pi/4} z)$

(a) $(w = z + (1 + 2i))$

If

$$[w = z + (1 + 2i),]$$

then

$$[u + iv = x + iy + 1 + 2i = (x + 1) + i(y + 2),]$$

and therefore

$$[u = x + 1, \quad v = y + 2.]$$

The line $(x = 0)$ is mapped into $(u = 1)$, $(y = 0)$ into $(v = 2)$, $(x = 2)$ into $(u = 3)$, and $(y = 1)$ into $(v = 3)$.

Thus (R) is mapped onto the rectangle in the (w) -plane bounded by

$$[u = 1, \quad u = 3, \quad v = 2, \quad v = 3.]$$

Example (1) (continued)

(b) $(w = e^{i/4} z)$

Let

$$[w = \sqrt{2} e^{i\pi/4} z.]$$

Then

$$[u + iv = \sqrt{2} (\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})(x + iy) = (1 + i)(x + iy)]$$
$$[u = (x - y) + i(x + y),]$$

so that

$$[u = x - y, \quad v = x + y.]$$

The line $(x = 0)$ is mapped into

$$[u = -y, \quad v = y,]$$

or

$$[u = -v.]$$

The line $(y = 0)$ is mapped into

$$[u = x, \quad v = x,]$$

or

$$[u = v.]$$

The line $(x = 2)$ is mapped into

$$[u = 2 - y, \text{quad } v = 2 + y,]$$

or

$$[u + v = 4.]$$

Finally, $(y = 1)$ is mapped into

$$[u = x - 1, \text{quad } v = x + 1,]$$

or

$$[v - u = 2.]$$

Thus the image (R') of the rectangle (R) is the parallelogram in the (w) -plane bounded by the lines

$$[u = v, \text{quad } u = -v, \text{quad } u + v = 4, \text{quad } v - u = 2.]$$

The mapping accomplishes a **rotation** of (R) through the angle $(\pi/4)$ and a **stretching** of lengths by the factor $(\sqrt{2})$.

Example (2)

Determine the region of the (w) -plane into which each of the following is mapped by the transformation

$$[w = z^2.]$$

(a) First quadrant of the (z) -plane

Let

$$[z = re^{i\theta}, \text{quad } w = \rho e^{i\phi}.]$$

Then

$$[\rho e^{i\phi} = z^2 e^{2i\theta},]$$

so that

$$[\rho = r^2, \text{quad } \phi = 2\theta.]$$

Thus points in the (z) -plane at $((r, \theta))$ are rotated through angle (2θ) and have their modulus squared.

All points in the first quadrant of the (z) -plane occupy the region

$$[0 < \theta < \frac{\pi}{2},]$$

and therefore they map into

$$[0 < \phi < \pi,]$$

the upper half-plane of the (w) -plane.

Example (2) (continued)

(b) Region bounded by $(x = 1)$, $(y = 1)$ and $(xy = 1)$

Since

$$[w = z^2 \quad \Longleftrightarrow \quad u + iv = (x + iy)^2 = x^2 - y^2 + 2ixy,]$$

we see that

$$[u = x^2 - y^2, \quad v = 2xy.]$$

The line $(x = 1)$ maps into

$$[u = 1 - y^2, \quad v = 2y,]$$

or

$$[u = 1 - \frac{v^2}{4}.]$$

The line $(y = 1)$ maps into

$$[u = x^2 - 1, \quad v = 2x,]$$

or

$$[u = \frac{v^2}{4} - 1.]$$

The curve $(xy = 1)$ can be written as $(y = 1/x)$ and maps into

$$[u = x^2 - \frac{1}{x^2}, \quad v = 2,]$$

so that

$$[v = 2]$$

is a horizontal line.

Eliminating x and y in this way, the boundary curves in the w -plane are

$$\left[u = 1 - \frac{v^2}{4}, \quad u = \frac{v^2}{4} - 1, \quad v = 2, \right]$$

and the corresponding region is the image of the curvilinear triangle bounded by $x = 1$, $y = 1$, and $xy = 1$ in the z -plane.

Physical Applications of Conformal Mapping

An important property of analytic functions is that they are **harmonic**; that is, their real and imaginary parts satisfy Laplace's equation.

Another important characteristic is that a **harmonic function remains harmonic under a conformal mapping**, which makes conformal mapping very useful in solving engineering problems.

Suppose, for example, that $\Phi(x,y)$ is a solution of

$$\left[\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0. \right. \tag{3.5}$$

Then, when an analytic function

$$\left[z = x + iy = f(u + iv) \right]$$

transforms Φ into a function of u and v , the transformed function $\Phi(u,v)$ will satisfy

$$\left[\frac{\partial^2 \Phi}{\partial u^2} + \frac{\partial^2 \Phi}{\partial v^2} = 0, \right. \tag{3.6}$$

wherever the mapping is conformal.

Flow of an Ideal Fluid Past a Circular Cylinder

As an illustration, consider a two-dimensional incompressible, irrotational flow of an ideal fluid past a circular cylinder of radius 1 whose axis is perpendicular to the xy -plane.

We choose coordinates so that the centre of the cylinder is at the origin and the undisturbed stream at infinity is parallel to the x -axis.

Let $\Phi(x,y)$ be the **velocity potential** and $\psi(x,y)$ the corresponding **stream function**.

Both Φ and ψ are harmonic and satisfy Laplace's equation outside the cylinder.

Boundary conditions:

1. The surface of the cylinder is a streamline, so $\psi = 0$ $\quad \text{on}$ $r = 1$.
2. The x -axis is also taken as a streamline, $\psi = 0$ $\quad \text{on}$ $y = 0$, so that the flow is symmetric about the x -axis and the fluid region is the upper half of the exterior of the circle $(r = 1)$.
3. At large distances from the cylinder the streamlines should be parallel (or nearly parallel) to the x -axis, corresponding to a uniform flow of constant speed c .

The complex potential of the flow will be constructed using a conformal mapping which transforms this region into a simpler one.

Hence the **complex potential** in the w -plane is

$$F = \Phi + i\psi = cu + icv = c w.$$

The analytic function

$$w = z + \frac{1}{z} \tag{3.14}$$

maps the upper half of the w -plane onto the region in the z -plane exterior to the unit circle $(r = 1)$ and above the x -axis.

By substituting $(x + iy)$ for (z) , we find that $(v = 0)$ when $(y = 0)$ or when $(x^2 + y^2 = 1)$.

Consequently, the u -axis maps onto the x -axis for $(x < -1)$ and $(x > 1)$, and onto the upper half of the circle $(x^2 + y^2 = 1)$ when $(-1 < x < 1)$.

Also, for very large values of $(|z|)$, we have $(w \approx z)$, so streamlines at large distances from the origin in the z -plane will be practically unaltered in their condition in the w -plane.

Using equation (3.14) for the transformation of $(F = cw)$, we get, with $(z = re^{i\theta})$,

$$F = c \left[\left(z + \frac{1}{z} \right) \right] = c \left[\left(r + \frac{1}{r} \right) \cos \theta + i \left(r - \frac{1}{r} \right) \sin \theta \right]. \tag{3.15}$$

The streamlines must lie outside the circle $(r = 1)$.

At large distances from the cylinder the streamlines should be parallel or nearly parallel to the x -axis.

The **complex potential** of the flow is given by the analytic function

$$F(z) = \Phi(x,y) + i\psi(x,y).$$

Taking advantage of symmetry, only the half of the (z) -plane above the (x) -axis needs to be considered.

The required solution is then the imaginary part of $(F(z))$ in the upper half of the (z) -plane.

Because of the shape of the boundary it is difficult to obtain the solution directly.

However, the solution is known for unobstructed uniform flow.

We assume that this uniform flow takes place in the **upper half of the (w) -plane**, parallel to the (u) -axis.

Then

$$V_1 = c, \quad V_2 = 0,$$

and since

$$V_1 = \frac{\partial \psi}{\partial v}, \quad V_2 = -\frac{\partial \psi}{\partial u},$$

the stream function in the (w) -plane is

$$\psi = c v,$$

where (c) is a real constant.

The velocity potential must therefore be

$$\Phi = c u$$

in order to satisfy the Cauchy–Riemann equations.

From (3.15) the **imaginary part** gives the streamlines:

$$\psi = c \left(r - \frac{1}{r} \right) \sin \theta = \text{constant}. \quad \text{tag{3.16}}$$

The streamline $(\psi = 0)$ consists of the circle $(r = 1)$ and the (x) -axis.

For very large values of (r) , the streamline is given approximately by

$$r \sin \theta = y = \text{constant},$$

so that far from the cylinder the flow approaches a uniform stream parallel to the (x) -axis.

The Complex Potential

From equations (3.7) and (3.8) it is seen that the velocity potential Φ is **harmonic**:

$$\left[\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0 \right]$$

Consequently, $\Phi(x,y)$ must have a harmonic conjugate $\psi(x,y)$ to which it is related by the Cauchy–Riemann equations.

The function $\psi(x,y)$ is called the **stream function**.

As Φ and ψ are harmonic conjugates, they define an analytic function

$$\left[F(z) = \Phi(x,y) + i\psi(x,y) \right] \tag{3.9}$$

This function is called the **complex potential** of the flow.

By differentiation,

$$\left[\frac{dF(z)}{dz} = \frac{\partial \Phi}{\partial x} + i \frac{\partial \psi}{\partial x} \right. \\ \left. \frac{\partial \psi}{\partial x} = \frac{\partial \Phi}{\partial x} - i \frac{\partial \Phi}{\partial y} = V_1 - iV_2 \right] \tag{3.10}$$

and the velocity is given by

$$\left[V = V_1 + iV_2 \right] \tag{3.11}$$

with magnitude

$$\left[|V| = \sqrt{V_1^2 + V_2^2} \right] \tag{3.12}$$

Equipotential Lines and Streamlines

The families of curves

$$\left[\Phi(x,y) = c, \quad \psi(x,y) = k \right] \tag{3.13}$$

where c and k are constants, are the **equipotential lines** and **streamlines** of the flow, respectively.

Continuing the procedure for solving boundary–value problems by conformal mapping:

1. **Solve the problem for the simpler region.**

2. Use the solution in (ii) to solve the given problem by employing the inverse mapping function.
-

Applications to Fluid Flow

The solution of many important problems in fluid flow is often achieved by complex-variable methods under the following assumptions.

1. **The fluid flow is two-dimensional.**
2. **The flow is steady**, i.e. the velocity of the fluid at any point depends only on the position $((x,y))$ and not on time.
3. **There exists a function (Φ) called the velocity potential**, such that

$$[V_1 = \frac{\partial \Phi}{\partial x}, \quad V_2 = \frac{\partial \Phi}{\partial y} \tag{3.7}]$$

1. **The fluid is incompressible**, i.e. the mass per unit volume of the fluid is constant.
Then

$$[\frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} = 0 \tag{3.8}]$$

which is the **equation of continuity**.

1. **The fluid is non-viscous.**

Dirichlet and Neumann Problems

Let (R) be a simply-connected region bounded by a simple closed curve (C) .

Two types of boundary-value problems are of great importance.

1. **Dirichlet problems** seek the determination of a function (Φ) which satisfies Laplace's equation in (R) and takes prescribed values on the boundary (C) .
2. **Neumann problems** seek the determination of a function (Φ) which satisfies Laplace's equation in (R) and whose normal derivative

$$[\frac{\partial \Phi}{\partial n}]$$

takes prescribed values on the boundary $\partial(C)$.

Solutions for Problems by Conformal Mapping

Boundary-value problems can be solved for any simply-connected region R which can be mapped conformally by an analytic function onto a simpler region.

The basic procedure is the following:

1. **Use the mapping function** to transform the boundary-value problem for the region R into a corresponding one for a simpler region.
2. **Solve the problem for the simpler region.**
3. **Use the solution** in step (2) to solve the given problem by employing the **inverse mapping function**.